DISTORTER: A CONTROLLABLE, REAL-TIME HARSH-VOICE EFFECT FOR SPEECH AND SINGING

Beomjin Park[1] **Kyogu Lee**[1,2,3]
[1]Department of Intelligence and Information, Seoul National University
[2]Interdisciplinary Program in Artificial Intelligence, Seoul National University
[3]Artificial Intelligence Institute, Seoul National University
{kc07030, kglee}@snu.ac.kr

ABSTRACT

Screams, growls and shrieks are central to metal singing but demanding to produce and hard to repeat exactly. We present *Distorter*, a playable effect with three technique-targeted modes for clean voice, using causal mel-to-mel converters trained on unpaired recordings with CycleGAN. Motivated by the physiology of these techniques, it adds an effective spectral residual around the retained harmonic structure and rescales every frame to the input's level, and an inference-time bias on that residual controls how far the harshness dominates. In automatic tests the converters leave distinguishable signatures on held-out voices, and suppression depth (residual dominance) increases monotonically with intensity for every input–mode pair. In isolated single-session tests all three rendering modes reach a real-time factor below one on an NVIDIA TITAN RTX, the streaming modes at configured algorithmic latencies of 42.67 and 442.67 ms. Listening tests and quantitative baseline/ablation comparisons remain future work.

1. INTRODUCTION

Harsh metal vocals are expressive but can be physically demanding to sustain, with performers reporting increased laryngeal pressure and transient throat discomfort after prolonged growling [1]. Their take-to-take variability further motivates a controllable post-production effect for clean vocals.

Existing tools include ANGUS, a real-time DSP effect for vocal roughness based on amplitude modulation and time-domain filtering [2], a neural voice editor that lets users redraw pitch, subharmonic and structural-noise curves [3], singing-technique conversion [4]. We instead formulate Distorter as a technique-targeted harsh-vocal effect—not speaker conversion or reference-based style transfer—with one intensity control, cycle-consistent non-parallel mel conversion [5, 6], and the input retained as an explicit carrier.

© B. Park and K. Lee. Licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0). **Attribution:** B. Park and K. Lee, "Distorter: A Controllable, Real-Time Harsh-Voice Effect for Speech and Singing", in *Extended Abstracts for the Late-Breaking Demo Session of the 27th Int. Society for Music Information Retrieval Conf.,* Abu Dhabi, UAE, 2026.

2. METHOD

Distorter operates on a 100-band log-mel magnitude $X \in \mathbb{R}^{F\times T}$ at 24 kHz. Its structure is motivated by reports that supraglottic sources can accompany vocal-fold phonation and produce technique-dependent spectral changes [7]. As a modeling prior, we approximate the net source effects accompanying harsh phonation under shared vocal-tract shaping with a residual anchored to the input's coarse spectral envelope, without estimating anatomical sources. For each $k \in \{\mathrm{scream}, \mathrm{growl}, \mathrm{shriek}\}$ the deployed clean-to-harsh generator $G_k : X \to Y$ is a past-only 2D ConvNeXt U-Net [8] predicting a bounded residual-level field $\ell_{\theta_k}(X)$; the converted mel follows

$$R_{\mathrm{eff}} = E_q(X) + \ell_{\theta_k}(X) + \Delta\ell, \tag{1}$$

$$\widetilde{Y} = \log(\exp X + \exp R_{\mathrm{eff}}), \tag{2}$$

$$\widehat{Y}_t = [G_k(X;\Delta\ell)]_t = \widetilde{Y}_t - \mathrm{LSE}_f(\widetilde{Y}_t) + \mathrm{LSE}_f(X_t). \tag{3}$$

applied per mel bin and frame, where $E_q(X)$ is the input's normalized eight-coefficient ($q = 8$) mel-cepstral envelope without its level term c_0 and LSE_f is the log-sum-exp over one frame's mel bins, whose last two terms restore the frame's level. The user sets $\Delta\ell$ at inference only; added after the bounded network output it can push R_{eff} past the range seen in training, and since (3) restores the frame level it acts as a change in how far the residual dominates rather than in the level of the take.

Keeping the input as an explicit carrier is intended to let the melody, content and timing of the performance carry through, while R_{eff}, added in the linear domain, puts energy around and between the retained harmonics.

For each target technique, G_k is trained from scratch jointly with its reverse generator F_k on unpaired clean and harsh recordings, using least-squares adversarial and cycle-consistency losses [5, 9], a spectrally normalized two-stream discriminator [10], and a custom feature-matching loss, inspired by [11], on the channel-wise mean and standard deviation of discriminator features. The selected runs use $\lambda_{\mathrm{adv}} = 2.5$, $\lambda_{\mathrm{cyc}} = 1$ and $\lambda_{\mathrm{FM}} = 20$.

Scream uses clean and vocal-fry scream recordings from EMO [12], together with clean and HARDCORE-SCREAM recordings from EMVD [13]. Growl and shriek each use EMVD's clean recordings, with DEATHGROWL and BLACKSHRIEK as their respective harsh targets.

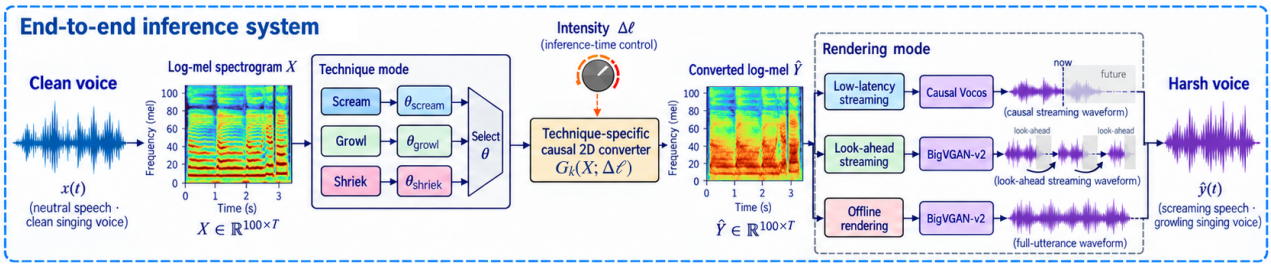


Figure 1: End-to-end inference. A selected causal converter adds a controllable effective residual; the same converted mel is rendered in low-latency, look-ahead, or offline mode.

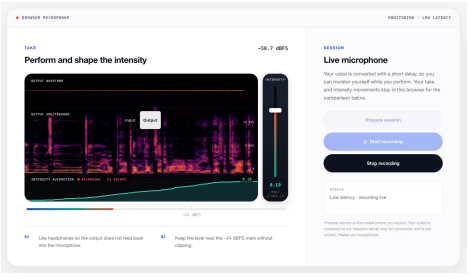


Figure 2: Demo interface with live output, recorded intensity automation, and session controls.

3. SYSTEM AND DEMONSTRATION

The 24-kHz browser demo [1] accepts microphone or audio-file input (Figures 1 and 2). Users monitor its output and control intensity live; the sample-indexed trajectory is replayed during offline re-rendering.

All paths share the converted mel: a causal Vocos-style decoder [14] for low-latency streaming and BigVGAN-v2 [15] for bounded-context or offline rendering. Both can introduce reconstruction error, while the causal decoder's reconstruction quality tended to vary more by input.

4. PRELIMINARY EVALUATION

All results are automatic; no listening test was run.

Mode separability. We classified converter outputs using 14 spectral/temporal-change features and leave-one-corpus-out logistic regression (Table 1). Each converter processed 616 VCTK speech [16] and GTSinger [17] and VocalSet [18] singing. This tests distinguishable output signatures, not perceived techniques. The growl-targeted mode separates best; 219/372 speech inputs converted by the shriek-targeted mode were classified as scream. In a four-way listening task on real recordings, not directly comparable, scream/shriek confusion was 22–24% [19].

Fidelity to the real techniques. The same features classify 36 real recordings at 41.7% (chance 33.3%; n.s.), which cannot establish resemblance to the real techniques. Without listening tests, we treat the modes as distinct, repeatable effects rather than exact technique reproductions.

Intensity control. Suppression depth is bottom-50% minus top-10% output-minus-input mel change, with bins ranked per input frame. All 1,848 curves increase at every $\Delta\ell$ step and are nearly linear (Table 1). Slopes decrease from about 2 to 1 dB/nat from dry to loud, so offsets are not perceptually uniform. Equation (3) preserves input frame levels within 1.44 dB for all frames and within 0.05 dB for 99.99% of frames within 70 dB of the take's peak.

Measurement	Result
Distinguishing the technique-targeted modes	
Separability: VCTK / GTSinger / VocalSet	78.4%, 88.1%, 75.8%
real recordings, same features	41.7% (n.s.)
Intensity control	
Suppression-depth response, -3 to $+1.5$ nats	$\rho = +1$, $r \geq +0.95$
peak-to-valley span, speech / singing	5–8 / 10–14 dB
frame-level matching, all / active	$\leq$1.44 / $\leq$0.05 dB

Table 1: Generated outputs: 616 inputs, 109 held-out voices, leave-one-corpus-out balanced accuracy (chance 33.3%; permutation $p = 0.002$, the 500-permutation floor); real row: 36 items, leave-one-singer-out. The VCTK/GTSinger/VocalSet speaker-cluster 95% bootstrap intervals are 76.3–80.6, 83.7–92.5 and 70.7–80.9%. Excluding inputs used in prior selection/evaluation panels changes pooled accuracy from 79.1% to 78.9% ($n = 508$). ρ and r are the worst itemwise Spearman and Pearson control correlations, respectively.

Throughput. On one idle NVIDIA TITAN RTX, worst RTFs (processing time/audio duration) are ≤ 0.63, ≤ 0.35 and ≤ 0.05 for low-latency, look-ahead and offline rendering. Configured streaming latencies are 42.67 and 442.67 ms; browser, network and device buffering and load are excluded.

Limits. In our listening, strongly harmonic inputs convert least convincingly because retained harmonicity opposes output harshness. Voice-transformation studies report F0 error, speaker similarity and ASR intelligibility [20]. Consistent with speaker-verification degradation under shouting [21], our ECAPA-TDNN EER rose from 11.1% on clean to 31.5% on paired harsh GT (nine lyrics). SwiftF0 [22] voiced coverage fell from 98.6% (clean) to 4.6% (harsh GT) at confidence 0.90; at 0.30 it reached 70.2%, but 44.0% of jointly voiced WORLD–SwiftF0 frames differed by $>$ 300 cents (27 vowels). Whisper-medium reached 69.2% WER on 20 real screams. A four-choice death-metal word-recognition study reported 65.9% for fans and 51.0% for non-fans [23]. Domain-validated listening tests, baselines and ablations remain future work.

[1] https://qjawls12.github.io/DISTORTER/

5. REFERENCES

[1] C. Eckers, D. Hütz, M. Kob, P. Murphy, D. Houben, and B. Lehnert, "Voice production in death metal singers," in *Proc. NAG/DAGA Int. Conf. on Acoustics*, Rotterdam, The Netherlands, 2009, pp. 1747–1750.

[2] M. Liuni, L. Ardaillon, L. Bonal, L. Seropian, and J.-J. Aucouturier, "ANGUS: Real-time manipulation of vocal roughness for emotional speech transformations," *arXiv preprint arXiv:2008.11241*, 2020.

[3] F. Bous, J. Morton, J. Byun, Y. Yu, J. Yang, H. Kim, and J. Lee, "A neural voice editing tool for transforming pitch, subharmonics, and structural noise," Late-Breaking/Demo, Int. Society for Music Information Retrieval Conf. (ISMIR), 2025.

[4] B. O'Connor, S. Dixon, and G. Fazekas, "Zero-shot singing technique conversion," in *Proc. Int. Symposium on Computer Music Multidisciplinary Research (CMMR)*, 2021, pp. 235–244.

[5] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, 2017, pp. 2242–2251.

[6] T. Kaneko, H. Kameoka, K. Tanaka, and N. Hojo, "CycleGAN-VC3: Examining and improving CycleGAN-VCs for mel-spectrogram conversion," in *Proc. Interspeech*, 2020, pp. 2017–2021.

[7] M. Aaen, J. McGlashan, N. Christoph, and C. Sadolin, "Extreme vocal effects distortion, growl, grunt, rattle, and creaking as measured by electroglottography and acoustics in 32 healthy professional singers," *Journal of Voice*, vol. 38, no. 3, pp. 795.e21–795.e35, 2024.

[8] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 11 976–11 986.

[9] X. Mao, Q. Li, H. Xie, R. Y. K. Lau, Z. Wang, and S. P. Smolley, "Least squares generative adversarial networks," in *Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, 2017, pp. 2794–2802.

[10] T. Miyato, T. Kataoka, M. Koyama, and Y. Yoshida, "Spectral normalization for generative adversarial networks," in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2018.

[11] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen, "Improved techniques for training GANs," in *Advances in Neural Information Processing Systems*, vol. 29, 2016, pp. 2234–2242.

[12] T.-C. Hsu, "EMO," Zenodo, version 0, 2026, dataset. [Online]. Available: https://doi.org/10.5281/zenodo.18356449

[13] M. Tailleur, J. Pinquier, L. Millot, C. Vogel, and M. Lagrange, "EMVD dataset: A dataset of extreme vocal distortion techniques used in heavy metal," in *Proc. Int. Conf. on Content-Based Multimedia Indexing (CBMI)*, 2024, pp. 1–5.

[14] H. Siuzdak, "Vocos: Closing the gap between time-domain and fourier-based neural vocoders for high-quality audio synthesis," in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2024.

[15] S.-G. Lee, W. Ping, B. Ginsburg, B. Catanzaro, and S. Yoon, "BigVGAN: A universal neural vocoder with large-scale training," in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2023.

[16] J. Yamagishi, C. Veaux, and K. MacDonald, "CSTR VCTK corpus: English multi-speaker corpus for CSTR voice cloning toolkit (version 0.92)," University of Edinburgh, Centre for Speech Technology Research (CSTR), 2019.

[17] Y. Zhang, C. Pan, W. Guo *et al.*, "GTSinger: A global multi-technique singing corpus with realistic music scores for all singing tasks," in *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*, vol. 37, 2024, pp. 1117–1140.

[18] J. Wilkins, P. Seetharaman, A. Wahl, and B. Pardo, "VocalSet: A singing voice dataset," in *Proc. Int. Society for Music Information Retrieval Conf. (ISMIR)*, 2018, pp. 468–474.

[19] X. Qiu and E. Parada-Cabaleiro, "Distorted realities: Classifying extreme vocals between harmony and noise — a machine–human evaluation of vocal confusion patterns," *Transactions of the International Society for Music Information Retrieval*, vol. 9, no. 1, pp. 309–328, 2026.

[20] B. Sisman, J. Yamagishi, S. King, and H. Li, "An overview of voice conversion and its challenges: From statistical modeling to deep learning," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 132–157, 2021.

[21] S. Prieto, A. Ortega, I. López-Espejo, and E. Lleida, "Shouted speech compensation for speaker verification robust to vocal effort conditions," in *Proc. Interspeech*, 2020, pp. 1511–1515.

[22] L. Nieradzik, "SwiftF0: Fast and accurate monophonic pitch detection," arXiv preprint arXiv:2508.18440, 2025.

[23] K. N. Olsen, W. F. Thompson, and I. Giblin, "Listener expertise enhances intelligibility of vocalizations in death metal music," *Music Perception*, vol. 35, no. 5, pp. 527–539, 2018.